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DEPARTMENT OF COMPUTER SYSTEMS AND MATHEMATICS (CSM)

BSC. INFORMATION SYSTEMS MANAGEMENT (BSC. ISM)

IS 224: WEB APPLICATION DEVELOPMENT

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PROJECT TITLE: E-LEARNING COURSE MANAGEMENT SYSTEM

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ABSTRACT

The E-Learning Course Management System was developed to address the challenges associated with conventional teaching and learning methods, including limited accessibility to educational resources, inefficient course administration, inadequate learner progress tracking, and poor communication between instructors and students. Traditional classroom-based and manually managed learning environments often struggle to support flexible, scalable, and technology-driven education, particularly in situations where remote learning is required. The increasing demand for digital learning platforms has created the need for a secure, efficient, and user-friendly Learning Management System (LMS) capable of supporting both instructors and learners in a centralized environment.

This project adopted the Agile Software Development Life Cycle (SDLC) together with the Object-Oriented Analysis and Design (OOAD) methodology to guide the planning, analysis, design, implementation, testing, and deployment phases. The system was developed using the Django web framework with Python as the backend programming language, PostgreSQL as the relational database management system, and modern web technologies for the user interface.

Comprehensive testing was conducted to evaluate the system's functionality, usability, reliability, and performance. The testing results demonstrated that the system successfully satisfied both the functional and non-functional requirements specified during the requirements analysis phase. The developed E-Learning Course Management System offers a secure, scalable, and efficient digital learning environment that enhances teaching effectiveness, improves learner engagement, and streamlines educational administration.

Future enhancements may include mobile application support, artificial intelligence-based personalized learning recommendations, live virtual classroom integration, gamification features, multilingual support, and cloud-based deployment to improve accessibility, scalability, and overall user experience.

List Of Abbreviations

ICT	Information and Communication Technologies
LMS	Learning Management System
OOAD	Object-Oriented Analysis and Design
SDD	System Design Document
SDLC	Software Development Life Cycle

Table of Contents

Abstract.....	i
List Of Abbreviations	ii
Table Of Contents	iii
List Of Tables	v
List Of Figures	v
CHAPTER ONE: INTRODUCTION	1
1.0 General Introduction	1
1.1 Statement of the Problem	1
1.2 Objectives.....	1
1.2.1 General Objective.....	1
1.2.2 Specific Objectives	2
1.3 Significance of the Study	2
CHAPTER TWO: METHODOLOGY.....	3
2.1 Introduction	3
2.2 Selected Methodology	3
CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN	4
3.1 Introduction	4
3.2 Requirement Analysis	4
3.2.1 Functional Requirements.....	4
3.2.1 Non-Functional Requirements.....	5
3.4 System Design.....	6
3.4.1 System Database Design	6
CHAPTER FOUR: IMPLEMENTATION AND TESTING.....	7
4.1 Introduction	7
4.2 Model Implementation and Core Functionality	7

4.3 Testing	9
CHAPTER FIVE: CHALLENGES, RECOMMENDATIONS, AND CONCLUSION	10
5.1 Challenges Faced	10
5.2 Recommendations	10
5.3 Further Improvements	10
5.4 Conclusion	10
REFERENCES.....	11

List of Tables

Table 2.1	Methodology Table
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List of Figures

Figure 3.1	Entity Relationship Diagram
Figure 4.1	User Login Interface
Figure 4.2	Course Management Interface
Figure 4.3	Learning Material Interface
Figure 4.4	Course Enrollment Interface
Figure 4.5	Progress Tracking Interface
Figure 4.6	Certificate Sample

CHAPTER ONE

INTRODUCTION

1.0 General Introduction

The rapid advancement of Information and Communication Technologies (ICT) has transformed the education sector by enabling the adoption of digital learning platforms that improve teaching and learning. Learning Management Systems (LMSs) allow educational institutions to manage courses, distribute learning materials, conduct online assessments, and monitor student progress through a centralized platform. As the demand for flexible and accessible education continues to grow, e-learning systems have become essential for improving educational quality and administrative efficiency (Laudon & Laudon, 2022).

The E-Learning Course Management System addresses these challenges by automating course management, student enrollment, content delivery, assessments, and progress monitoring. (Al-Fraihat et al., 2020).

1.1 Statement of the problem

Many educational institutions continue to face challenges in managing teaching and learning activities due to dependence on traditional classroom methods and disconnected digital solutions. These challenges affect both instructors and learners by reducing the effectiveness of course management and limiting opportunities for flexible and continuous learning.

The growing demand for accessible and technology-driven education has created the need for a reliable E-Learning Course Management System that can provide a centralized platform for managing academic activities. Therefore, this project focuses on developing a system that supports course management, student enrollment, learning material distribution, online assessments, progress tracking, and communication between users.

1.2 Objectives

1.2.1 General objective.

The general objective of this project is to design and develop an E-Learning Course Management System that improves learning accessibility, enhance administrative efficiency examination results

1.2.2 Specific objectives.

- i. To define the functional and non-functional requirements of the E-Learning Course Management System based on the system's intended operations.
- ii. To design the structure and architecture of the E-Learning Course Management System.
- iii. To implement the E-Learning Course Management System using appropriate technologies and tools.
- iv. To test and evaluate the performance, accuracy, and reliability of the developed E-Learning Course Management System

CHAPTER TWO

METHODOLOGY

2.1 Introduction

Methodology refers to the systematic and theoretical analysis of the methods applied to a field of study. It comprises the theoretical underpinnings and the practical tools and procedures used in research (Kothari, 2004).

2.2 Selected Methodology

The E-Learning Course Management System was developed using **Agile Software Development methodology** combined with **Object-Oriented Analysis and Design (OOAD)**. Agile provided an iterative approach for continuous development, testing, and improvement, while OOAD was used to design system components as objects such as User, Course, and Assessment. The system was implemented through short development cycles (sprints), allowing features to be improved based on feedback and changing requirements (Pressman & Maxim, 2020). Testing was conducted throughout development to ensure system functionality, reliability, and usability according to user needs (Sommerville, 2021).

2.3 Methodology Table

Table 2-1: Methodology Table

S/N	Specific Objective	Methodology	Tools Used	Deliverable
1	To design the E-Learning Course Management System	Object-Oriented Analysis and Design (OOAD)	Draw.io	System Design Document (SDD)
2	To implement the E-Learning Course Management System	Agile Development Model	Python, VS Code, PostgreSQL	E-Learning Course Management System
3	To test and validate the E-Learning Course Management System	Alpha Testing Manual Functional Testing	Computers, Checklists, Sample Data	User and developer Feedback

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design of the E-Learning Course Management System. It explains how the system requirements were analyzed and translated into a structured system design. The chapter also describes the system architecture and modeling tools used to represent system functionality and data relationships.

3.2 Requirement Analysis

3.2.1 Functional Requirements

Functional requirements describe what the system must do in order to operate effectively.

- i. **User Registration:** The system must allow administrators to create and manage user accounts for instructors, students, and administrators. This ensures that users are properly registered before accessing the platform.
- ii. **User Authentication (Login and Logout):** The system must verify user credentials during login and provide secure logout functionality. This ensures that only authorized users can access the system resources.
- iii. **Role-Based Access Control:** The system must control access based on user roles (Administrator, Instructor, and Student). Each user should only access functionalities related to their assigned responsibilities.
- iv. **Course Management:** The system must allow instructors and administrators to create, update, and manage courses. Course information such as title, description, category, and learning resources should be properly stored and maintained.
- v. **Student Enrollment Management:** The system must allow students to enroll in available courses and enable instructors or administrators to monitor enrolled learners.
- vi. **Learning Material Management:** The system must allow instructors to upload, update, and manage learning materials such as documents, videos, and other educational resources for students.

- vii. **Online Assessment Management:** The system must allow instructors to create and manage quizzes, assignments, and assessments. Students should be able to complete assessments through the platform.
- viii. **Automatic Assessment Evaluation:** The system should automatically evaluate objective assessments such as quizzes and provide results to students. This reduces manual grading efforts and improves efficiency.
- ix. **Progress Tracking:** The system must monitor student learning activities, course completion status, and assessment performance to help instructors evaluate learner progress.
- x. **Certificate Generation:** The system must generate completion certificates for students who successfully complete required courses and assessments.

3.2.2 Non-Functional Requirements

Non-functional requirements describe how the system should perform and the quality standards it must meet. They include

- i. **Security:** The system must protect user data through secure authentication, password encryption, and role-based access control.
- ii. **Usability:** The system must provide a simple and user-friendly interface that allows users to easily access and manage learning activities.
- iii. **Performance:** The system should respond quickly to user requests, including accessing courses, materials, and assessments.
- iv. **Reliability:** The system must operate consistently with minimal errors and ensure successful completion of user activities.
- v. **Scalability:** The system should support future expansion by accommodating more users, courses, and additional features.
- vi. **Maintainability:** The system should have a modular design that allows easy updates, modifications, and maintenance.
- vii. **Availability and Compatibility:** The system should provide continuous access to learning resources and services. The system should work effectively across different browsers and devices.

3.4 System Design

3.4.1 System Database Design

The E-Learning Course Management System uses a relational database design implemented using **PostgreSQL**. The database consists of related tables such as Users, Courses, Enrollments, Learning Materials, Assessments, Progress, and Certificates, which are connected through primary and foreign keys to maintain data relationships and integrity. The design ensures efficient storage, retrieval, and management of learning information while supporting system security and scalability.

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**E - LEARNING COURSE MANAGEMENT SYSTEM
ERD DIAGRAM**

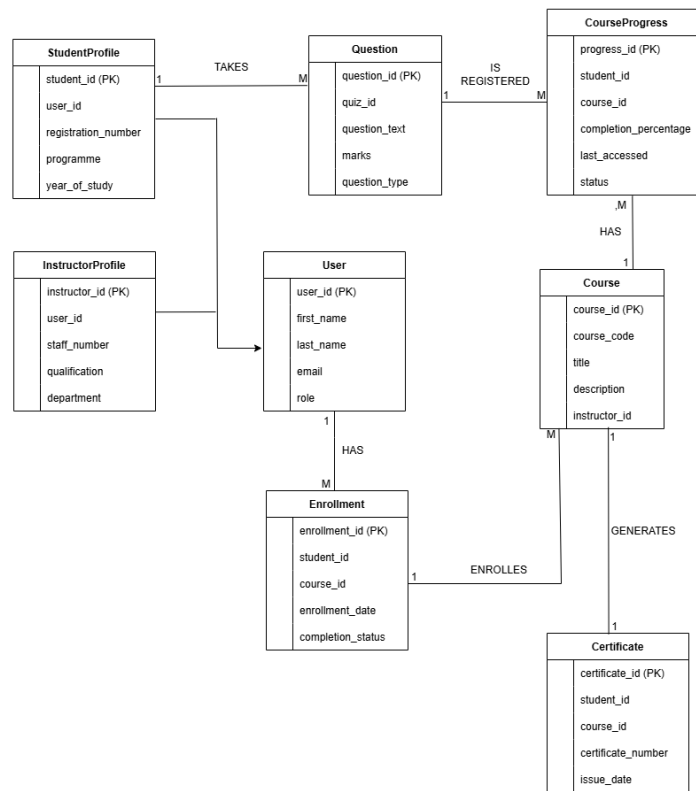


Figure 3.1 Entity Relationship Diagram

CHAPTER FOUR

IMPLEMENTATION AND TESTING

4.1 Introduction

This chapter presents the implementation and testing of the E-Learning Course Management System. It describes the development of the main system modules and explains how each component functions to support examination result processing.

4.2 Model Implementation and Core Functionality

The E-Learning Course Management System was implemented using the designed models, including User, Student, Instructor, Course, Enrollment, Learning Material, Assessment, Progress, and Certificate. The implemented functionalities demonstrate how users interact with the system to perform learning and management activities.

4.2.1 User Authentication

The system provides login functionality to verify users and allow secure access based on their roles.

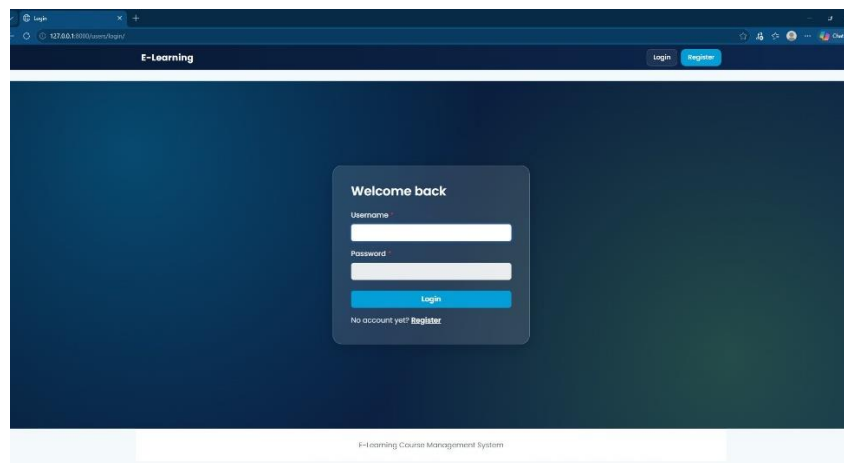
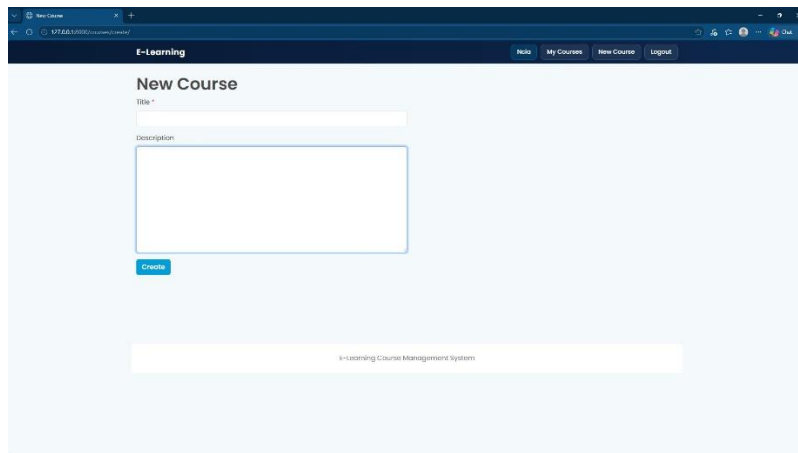


Figure 4.1: User Login Interface

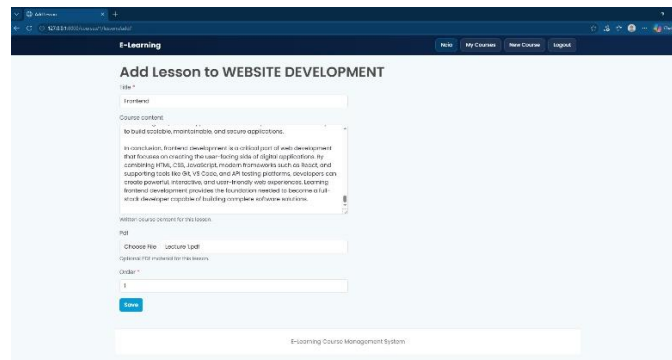
4.2.2 Course Management

The system allows instructors to create and manage courses available for students.



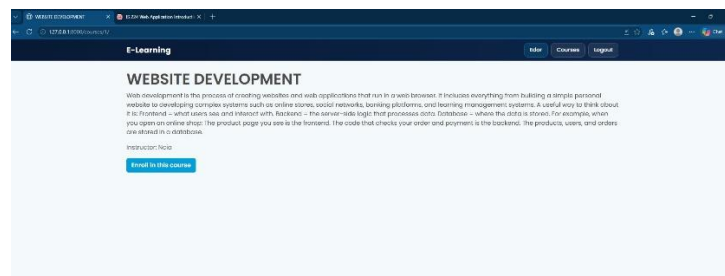
4.2.3 Learning Material Management

The system enables instructors to upload learning materials and allows students to access course resources.



4.2.4 Student Enrollment

The system allows students to enroll in courses and access their registered courses.



4.2.5 Progress Tracking

The system monitors student learning progress and course completion status.



Figure 4.5: Progress Tracking Interface

4.2.6 Certificate Generation

The system generates certificates for students who successfully complete required courses.

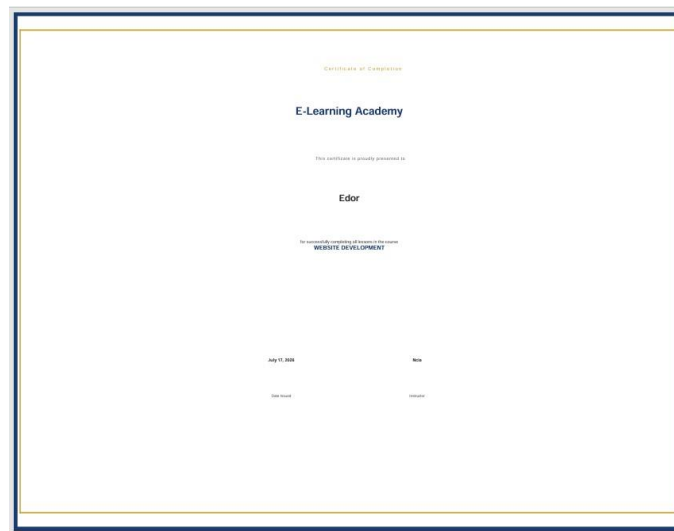


Figure 4.6: Certificate Sample

4.3 Testing

The system was tested using two methods: Alpha Testing and Manual Testing.

Alpha testing was conducted internally by the developers to verify that the system modules functioned correctly before deployment (Sommerville, 2011).

Manual testing was also performed by executing test cases to validate system functionality, user interactions, and ensure that features operated according to the requirements (Oates, 2006).

CHAPTER FIVE

CHALLENGES, RECOMMENDATIONS, AND CONCLUSION

5.1 Challenges Faced

During the development of the E-Learning Course Management System, several challenges were encountered. Limited time affected extensive testing and collection of feedback from different users. Technical challenges were also experienced during system integration, database management, and implementation of different modules such as authentication, course management, and assessments. Additionally, ensuring system usability and compatibility across different devices required extra effort.

5.2 Recommendations

To improve the system, enhanced security features such as stronger authentication, data backup, and activity monitoring should be implemented. The system should also be deployed on cloud infrastructure to improve accessibility, scalability, and availability for a larger number of users. User training is recommended to ensure effective utilization of the system functionalities.

5.3 Further Improvements

Future improvements may include developing a mobile application, integrating live virtual classrooms, adding artificial intelligence-based learning recommendations, and introducing advanced analytics for learner performance tracking. Additional features such as notifications, discussion forums, and multilingual support can also be integrated to enhance the learning experience.

5.4 Conclusion

The E-Learning Course Management System was successfully developed to address challenges associated with traditional learning management approaches. The system provides essential functionalities including user authentication, course management, learning material access, enrollment, assessments, and progress tracking, improving efficiency and accessibility in digital learning. Despite some technical and time limitations, the system achieved its objectives and provides a foundation for future enhancements and expansion.

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